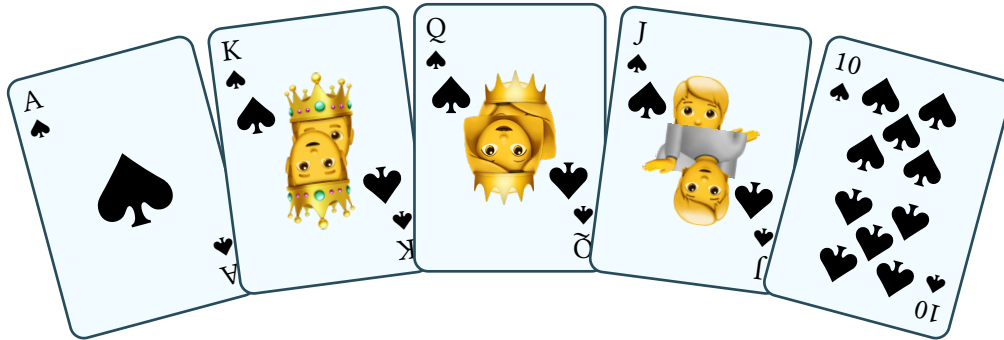


Probability Education: Hand Probabilities

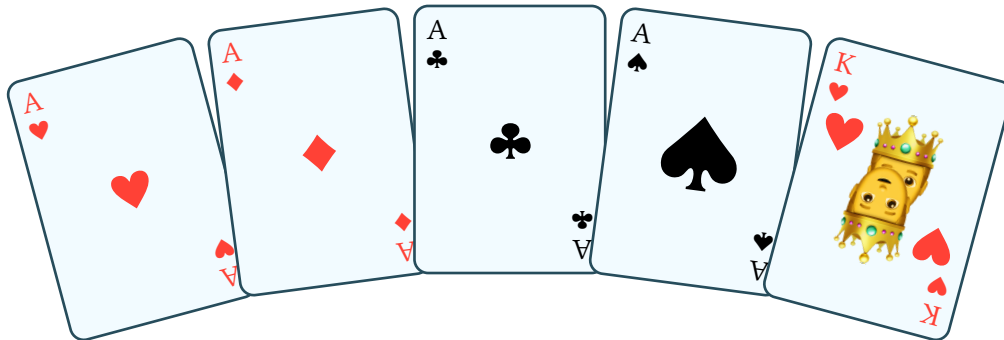
Game State

Player Hands

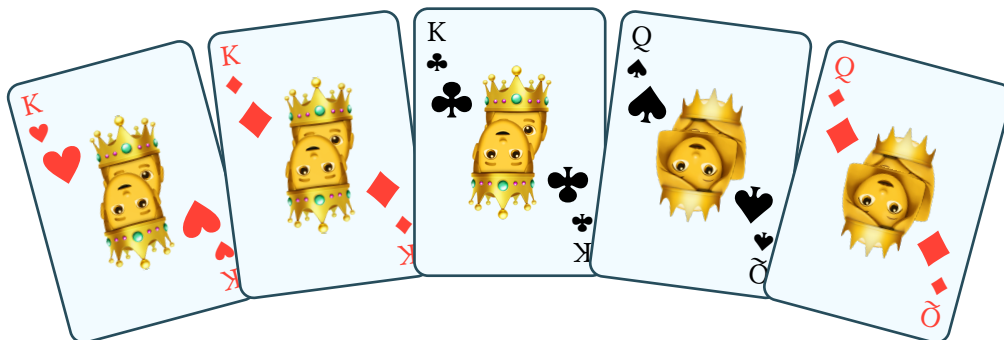
Royal Flush (Rarest):



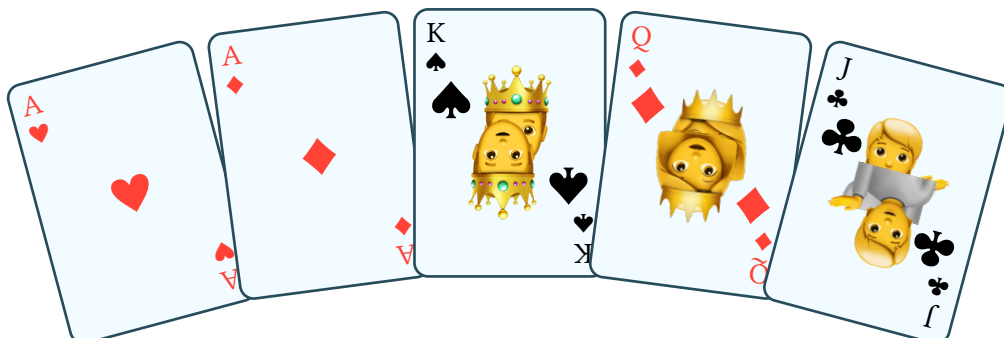
Four of a Kind:



Full House:



One Pair (Most Common):



Additional Content

Probability Education: Hand Probabilities

Concept Overview

This lesson demonstrates the probability of different poker hands. Each hand type has a specific probability based on mathematics.

Key Insight

Probability isn't about luck—it's about mathematics. The cards don't care about your feelings. They follow the laws of probability.

Visual Examples

Royal Flush (Rarest)

The rarest hand. Only 4 possible combinations out of 2,598,960 total hands.

****Probability**:** 1 in 649,739 (0.000154%)

Four of a Kind

Four cards of the same rank. Much more common than a royal flush.

****Probability**:** 1 in 4,164 (0.024%)

Full House

Three of a kind plus a pair. A strong but achievable hand.

****Probability**:** 1 in 693 (0.144%)

One Pair (Most Common)

Just one pair. The most common hand type in poker.

****Probability**:** 1 in 1.37 (42.26%)

The Mathematics

$$\text{Probability} = (\text{Number of favorable outcomes}) / (\text{Total number of outcomes})$$

For poker hands:

- Total 5-card hands from 52 cards: 2,598,960
- Each hand type has a specific count
- $\text{Probability} = \text{Count} / 2,598,960$